A parabolic counterexample to a conjecture on chaotic polynomial shifts

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Abstract

For every $1 < p < \infty$, the polynomial $P(x)_j = x_{j+1} + x_{j+1}^2$ on complex ℓ^p is topologically mixing and has dense periodic points. Its scalar polynomial has derivative one at zero. This disproves the proposed necessity of a repelling fixed point in MathDB 394128. The proof uses an explicit inverse branch and an elementary five-step covering argument.

Task 1. Resolve the proposed characterization. The conjecture on printed page 2241 of [1] asks whether, for a complex polynomial q of degree greater than one with $q(0) = 0$, the map

$$P : \ell^p \longrightarrow \ell^p, \qquad P(x)_j = q(x_{j+1}),$$

is Devaney chaotic exactly when zero is a repelling fixed point of q. Here $1 \le p < \infty$, and the source defines Devaney chaos as topological transitivity together with density of periodic points.

Theorem 1. *Let $q(z) = z + z^2$. For each $1 < p < \infty$, the associated map P on complex ℓ^p is a continuous polynomial, is topologically mixing, has dense periodic points, and has sensitive dependence on initial conditions. Nevertheless $q'(0) = 1$, so zero is not repelling. In particular $p = 2$ gives a counterexample to the conjecture.*

Throughout, superscripts on q and on the inverse branch below denote iteration, and $H = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$.

Lemma 1. *For $q(z) = z + z^2$, one has $q^5(H) = \mathbb{C}$.*

Proof. First,

$$q(H) = \{s + it : s + t^2 > 0\}. \tag{1}$$

Indeed write a square root of $w + 1/4$ as $A + iB$, choosing $A \ge 0$. The two preimages of w are $-1/2 \pm (A + iB)$. Moreover,

$$\operatorname{Re} w + (\operatorname{Im} w)^2 = (A^2 - 1/4)(1 + 4B^2).$$

Thus a preimage belongs to H precisely when $A > 1/2$, proving (1).

If $w \notin q^2(H)$, neither of its preimages belongs to $q(H)$. Applying (1) to these preimages gives $A + B^2 \le 1/2$. Consequently $0 \le A \le 1/2$ and

$$|w + 1/4| = A^2 + B^2 \le A^2 - A + 1/2 \le 1/2, \qquad |w| \le 3/4.$$

We have proved

$$\{w : |w| > 3/4\} \subset q^2(H). \tag{2}$$

Put

$$v = \frac{-1+i}{2}, \qquad c = \frac{-1-\sqrt{-1+2i}}{2},$$

using the principal square root. Then $q(c) = v$, $q(v) = -1/2$, and

$$\operatorname{Re}\sqrt{-1+2i} = \sqrt{\frac{\sqrt{5}-1}{2}} > 1/2.$$

Hence $\operatorname{Re} c < -3/4$, while $(q^3)'(c) = 0$. For every $w \in \mathbb{C}$, some root u of $q^3(z) - w$ must therefore satisfy $|u| > 3/4$. To justify this last assertion directly, if all roots $r_1, \dots, r_8$, counted with multiplicity, lay in the closed disk of radius $3/4$, the logarithmic derivative at c would give

$$0 = \frac{c(q^3)'(c)}{q^3(c) - w} = \sum_{\nu=1}^{8} \frac{1}{1 - r_\nu/c}.$$

The denominator on the left is nonzero under that assumption. Every summand on the right has positive real part, since $|r_\nu/c| < 1$; this is a contradiction. By (2), the resulting root has the form $u = q^2(h)$, $h \in H$, and then $q^5(h) = w$. □

Lemma 2. *The map*

$$g(z) = \frac{-1+\sqrt{1+4z}}{2} = \frac{2z}{1+\sqrt{1+4z}}, \qquad z \in H,$$

with the principal square root, maps H into itself and satisfies $q \circ g = \mathrm{id}_H$. For $h \in H$ and $k \geq 1$,

$$|g^k(h)| \leq \frac{(1+|h|)^2}{k}. \tag{3}$$

Proof. For $z \in H$, the real part of $\sqrt{1+4z}$ is greater than one: its square is $(|1+4z|+\operatorname{Re}(1+4z))/2 > 1$. Thus $g(z) \in H$, and direct substitution proves the inverse identity. Set $a_k = g^k(h)$ and $M = |h|$. Since $a_k = a_{k+1}(1+a_{k+1})$, the moduli $|a_k|$ decrease and are at most M. Also

$$\frac{1}{a_{k+1}} - \frac{1}{a_k} = \frac{1}{1+a_{k+1}}, \qquad \operatorname{Re}\frac{1}{1+a_{k+1}} = \frac{1+\operatorname{Re} a_{k+1}}{|1+a_{k+1}|^2} \geq \frac{1}{(1+M)^2}.$$

Summing gives $\operatorname{Re}(1/a_k) \geq \operatorname{Re}(1/h) + k/(1+M)^2 > k/(1+M)^2$. Since $|1/a_k| \geq \operatorname{Re}(1/a_k)$, this proves (3). □

Proof of Theorem 1. The coordinatewise product is a continuous bilinear map on ℓ^p, because

$$\|(x_j y_j)_j\|_p \leq \|x\|_\infty \|y\|_p \leq \|x\|_p \|y\|_p.$$

Composing its associated quadratic polynomial with the bounded backward shift, and adding that shift, gives precisely P. Thus P is a continuous polynomial of degree two and is defined on all of ℓ^p. Induction yields

$$(P^N x)_j = q^N(x_{j+N}). \tag{4}$$

For mixing, let U, V be nonempty open sets and choose finitely supported $a \in U$, $b \in V$. Take K containing both supports. For each j with $b_j \neq 0$, Lemma 1 supplies $h_j \in H$ with $q^5(h_j) = b_j$; put $C_j = (1+|h_j|)^2$. For every integer $N > K$, $N \geq 6$, set

$$x^{(N)} = a + \sum_{\{j : b_j \neq 0\}} g^{N-5}(h_j) e_{N+j}.$$

The appended coordinates are disjoint from the support of a. The inverse identity and (4) give $P^N(x^{(N)}) = b$, and

$$\|x^{(N)} - a\|_p \le \frac{\left(\sum_{\{j:b_j\neq 0\}} C_j^p\right)^{1/p}}{N-5} \longrightarrow 0.$$

Thus $P^N(U) \cap V \neq \varnothing$ for every sufficiently large N. This proves mixing, in fact for every $1 \le p < \infty$. If hypercyclicity is used in the definition, it follows as well: for a countable base (V_m), each $\bigcup_{n\ge 0} P^{-n}(V_m)$ is open and dense, so the Baire theorem provides a point whose orbit meets every V_m.

For density of periodic points, let a be supported in $\{1, \dots, K\}$. For each nonzero a_j, choose $h_j \in H$ with $q^5(h_j) = a_j$, and again let $C_j = (1 + |h_j|)^2$. Given an integer $N > K$, $N \ge 10$, define

$$y_j = a_j \quad (1 \le j \le K), \qquad y_{j+kN} = g^{kN-5}(h_j) \quad (a_j \neq 0,\ k \ge 1),$$

and set every remaining coordinate to zero. Residue classes modulo N do not overlap. As $kN - 5 \ge kN/2$,

$$\|y - a\|_p^p \le \sum_{\{j:a_j\neq 0\}} \sum_{k=1}^{\infty} \frac{C_j^p}{(kN-5)^p} \le \left(\frac{2}{N}\right)^p \left(\sum_{k=1}^{\infty} k^{-p}\right) \sum_{\{j:a_j\neq 0\}} C_j^p. \tag{5}$$

For $p > 1$ this is finite and tends to zero as $N \to \infty$. The first block satisfies $q^N(y_{j+N}) = q^5(h_j) = a_j$. In every subsequent nonzero residue class,

$$q^N\big(g^{(k+1)N-5}(h_j)\big) = g^{kN-5}(h_j) \quad (k \ge 1).$$

Zero residue classes remain zero, so (4) proves $P^N y = y$. Finitely supported vectors are dense; hence periodic points are dense.

For completeness, sensitivity has the uniform constant one. Given $x \in \ell^p$ and $\varepsilon > 0$, choose $0 < b < \varepsilon/4$. The positive real iterates $q^n(b)$ tend to infinity: they increase, and a finite limit would be a positive fixed point of q, which does not exist. Choose j so large that $|x_j| < \varepsilon/4$ and $q^{j-1}(b) > 2$. Replacing coordinate j of x by zero or by b gives two points within distance ε of x. After $j - 1$ iterates their first coordinates differ by more than two. The triangle inequality puts at least one more than distance one from $P^{j-1}x$. Finally $q'(0) = 1$, completing the counterexample. □

The summability step (5) requires $p > 1$. No assertion about dense periodic points on ℓ^1 is made here. The original equivalence is already false on ℓ^2.

Task 2. Determine the parabolic summability threshold. For $q_m(z) = z + z^{m+1}$, $m \ge 1$, determine exactly which exponents $1 \le p < \infty$ give dense periodic points for the associated shift. Relate the threshold to the decay of inverse iterates near zero, including the borderline exponent.

Task 3. Analyze irrationally indifferent fixed points. For polynomials with $q(0) = 0$ and $q'(0) = e^{2\pi i\alpha}$, α irrational, determine which local behaviors at zero allow Devaney chaos on ℓ^p. Treat the Siegel and Cremer cases separately.

Task 4. Investigate stronger recurrence. Determine whether the shift in Theorem 1 is frequently hypercyclic and whether it admits a fully supported invariant Borel probability measure that is mixing.

Task 5. Extend the construction to weighted spaces. Find necessary and sufficient weight conditions for this parabolic polynomial shift to be well defined, mixing, and Devaney chaotic on weighted ℓ^p and Köthe echelon spaces.

References

[1] F. Martínez-Giménez (joint work with A. Peris), Hypercyclic and chaotic polynomials on Fréchet spaces, in *Mini-Workshop: Hypercyclicity and Linear Chaos*, Oberwolfach Report 37/2006, pp. 2238–2241. Publisher report.